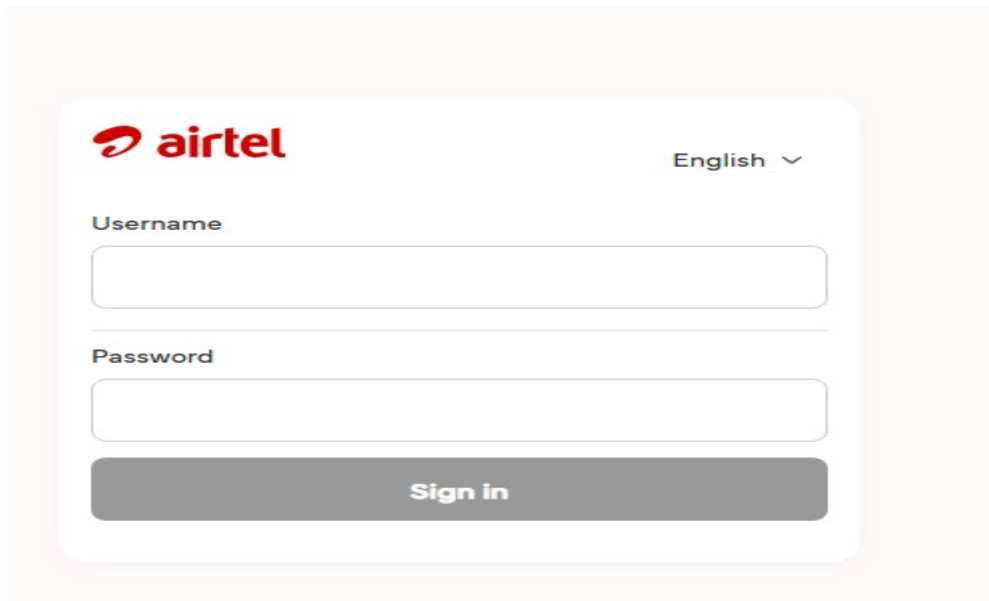


1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

 Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

IPv4 Address. : 192.168.1.88

Subnet Mask : 255.255.255.0

Default Gateway : fe80::a291:caff:fe28:3381%15 192.168.1.1

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

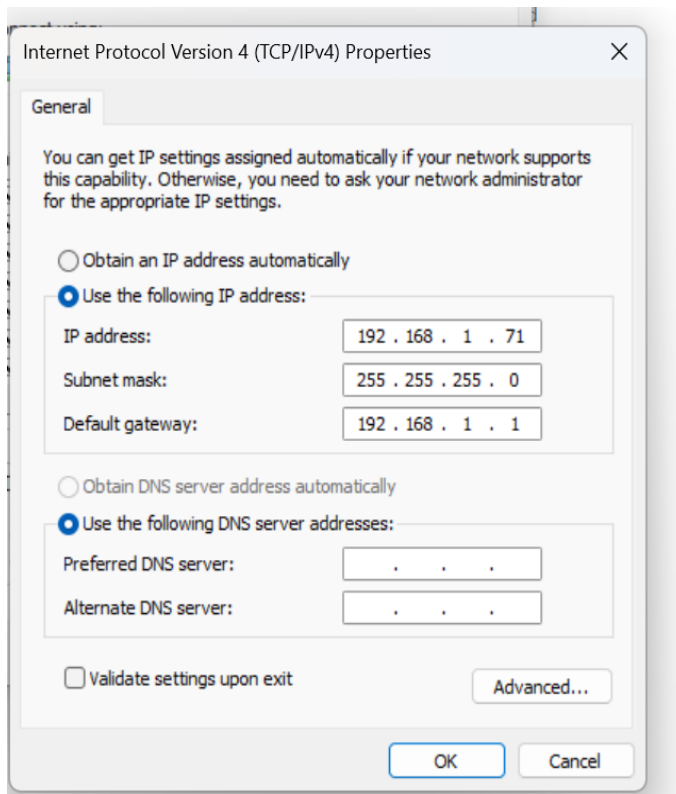
 Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

```

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2401:4900:88b7:b5d5:9018:1018:22c2:1714
Temporary IPv6 Address. . . . . : 2401:4900:88b7:b5d5:d85d:2254:3d2e:6498
Link-local IPv6 Address . . . . . : fe80::eec0:9248:fb84:eee8%15
IPv4 Address. . . . . : 192.168.1.71
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::a291:caff:fe28:3381%15
                             192.168.1.1

```



4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

If two devices are assigned the same IP address, an **IP address conflict** occurs. This can cause one or both devices to lose network connectivity, experience unstable internet access, or be unable to communicate properly with other devices on the network. DHCP prevents this by assigning unique IP addresses to each connected device.